Thinking in Parallel

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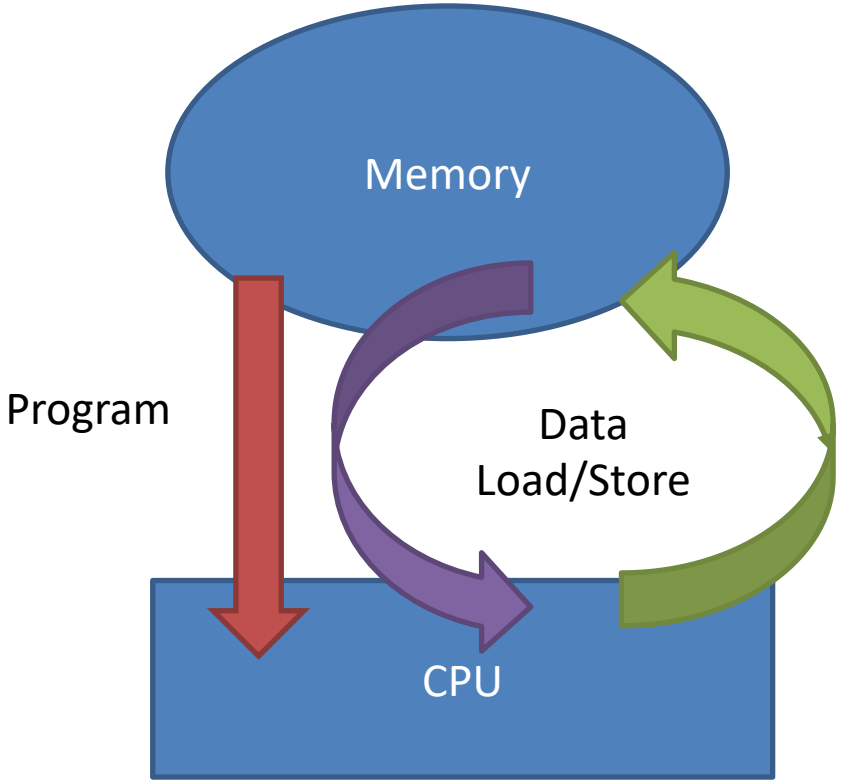
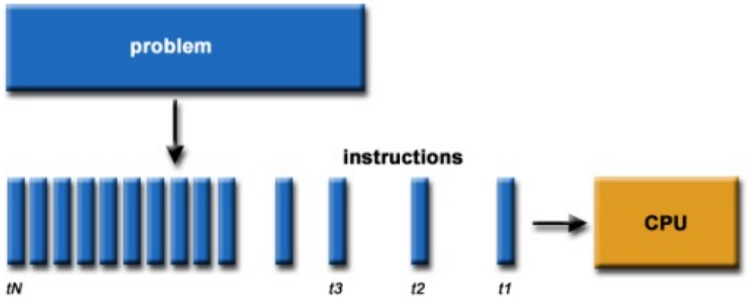
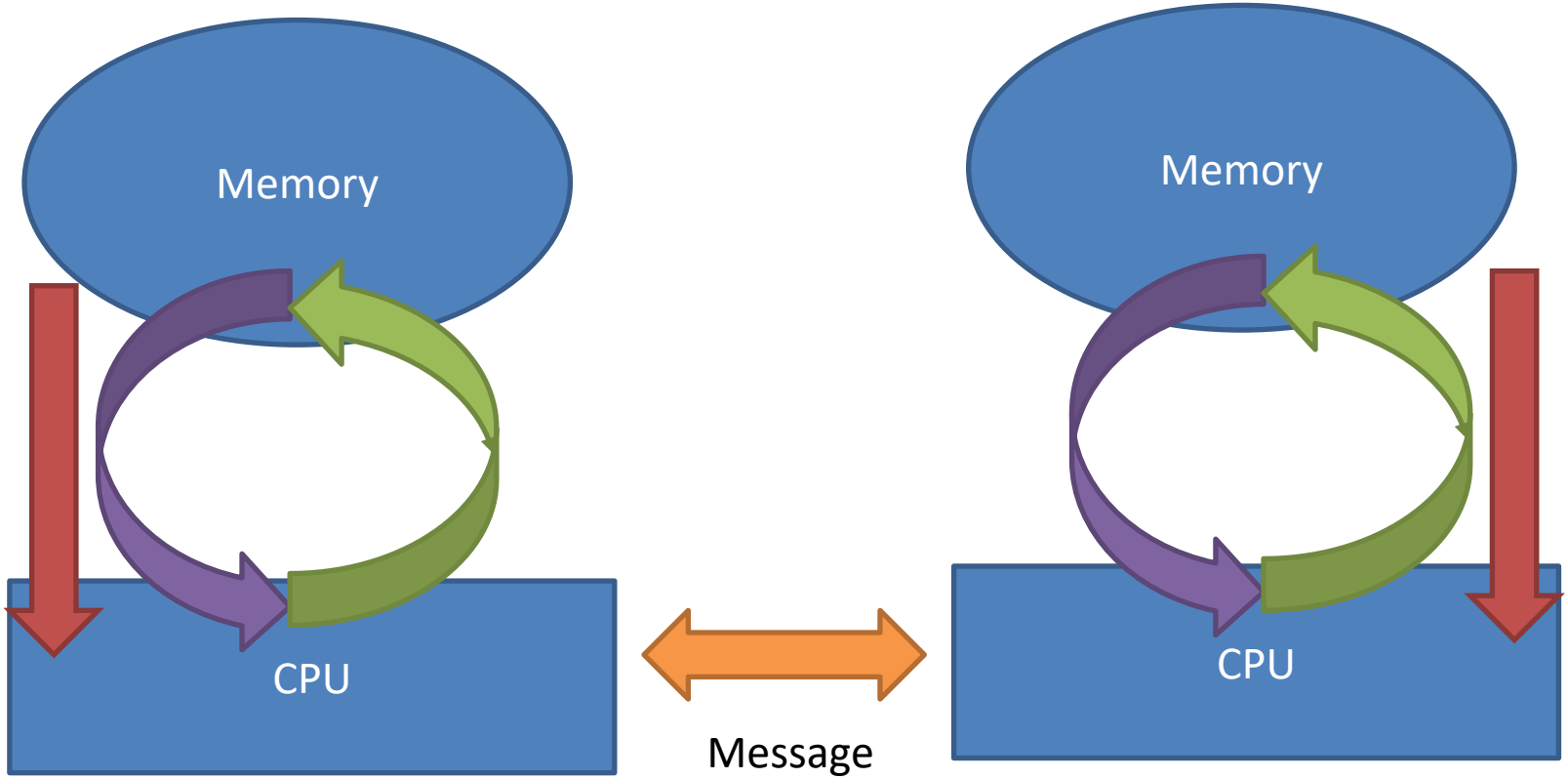
Serial Programming



A problem is broken into a discrete series of instructions.
Instructions are executed one after another.
Only one instruction may execute at any moment in time.

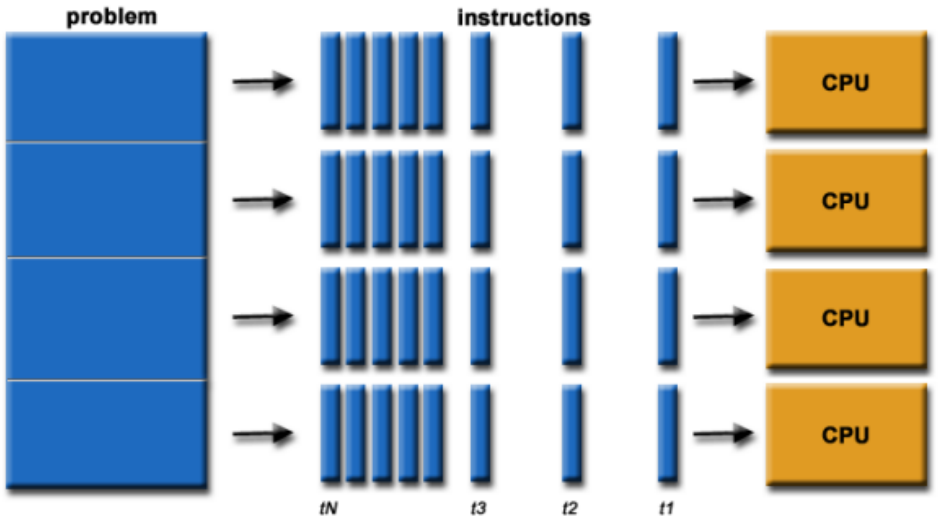


Parallel Programming



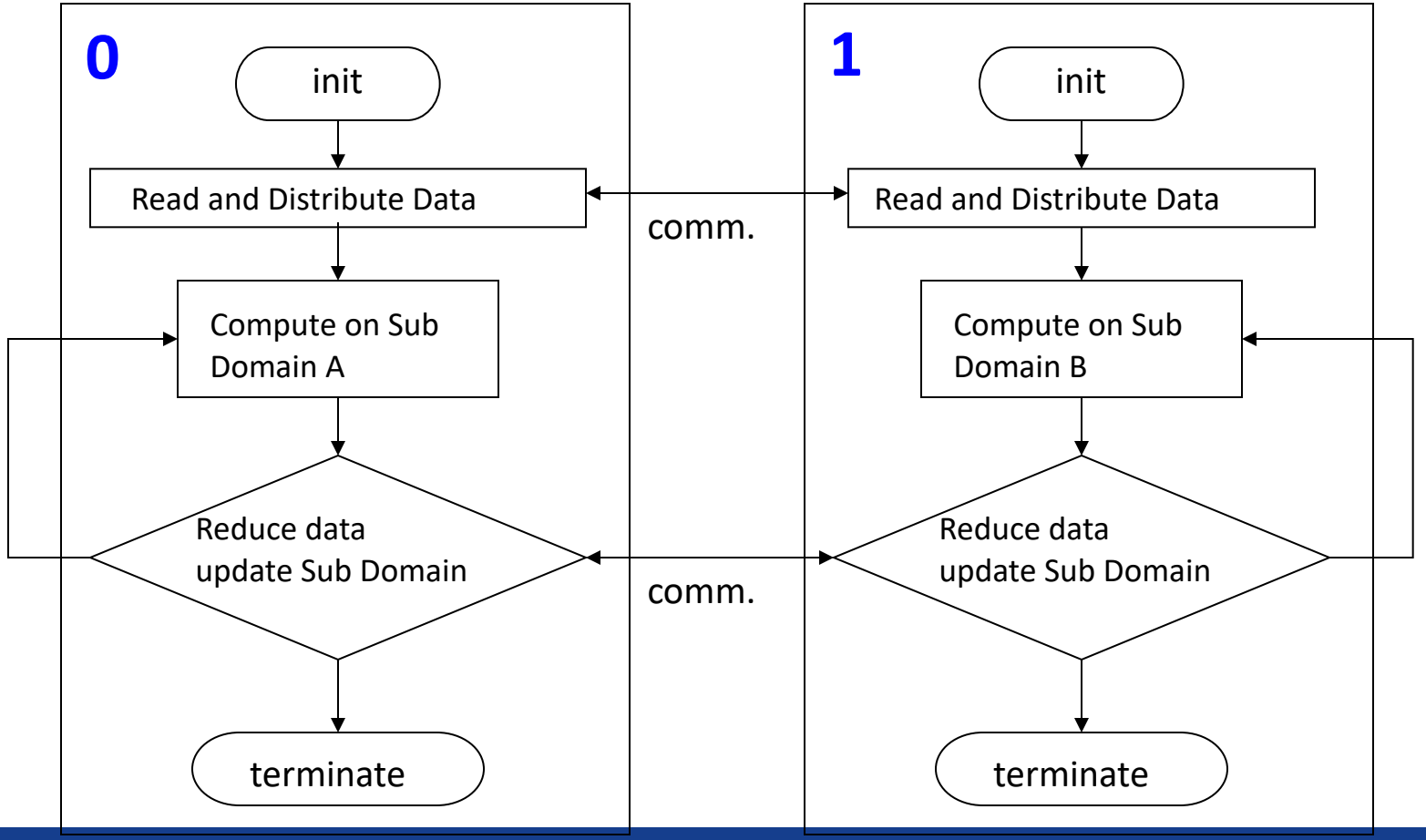
Concurrency

The first step in developing a parallel algorithm is to decompose the problem into tasks that can be executed concurrently



- A problem is broken into discrete parts that can be solved concurrently
- Each part is further broken down to a series of instructions
- Instructions from each part execute simultaneously on different processors
- An overall control / coordination mechanism is employed

What is a Parallel Program



Fundamental Steps of Parallel Design

- Identify portions of the work that can be performed concurrently
- Mapping the concurrent pieces of work onto multiple processes running in parallel
- Distributing the input, output and intermediate data associated within the program
- Managing accesses to data shared by multiple processors
- Synchronizing the processors at various stages of the parallel program execution

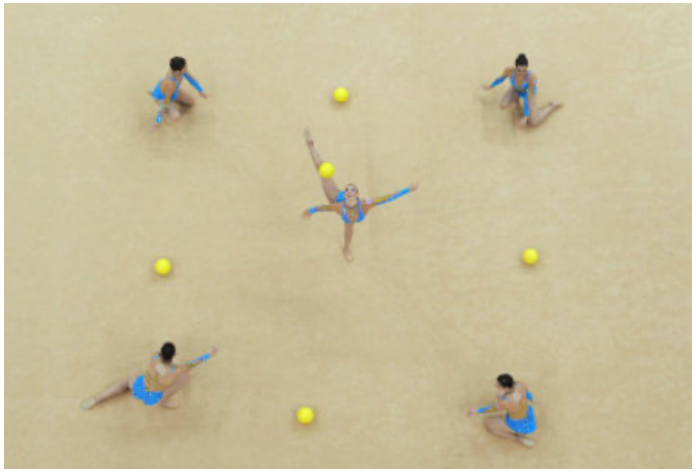
Type of Parallelism

- **Functional (or task) parallelism**: different people are performing different task at the same time

- **Data Parallelism**: different people are performing the same task, but on different equivalent and independent objects

Process Interactions

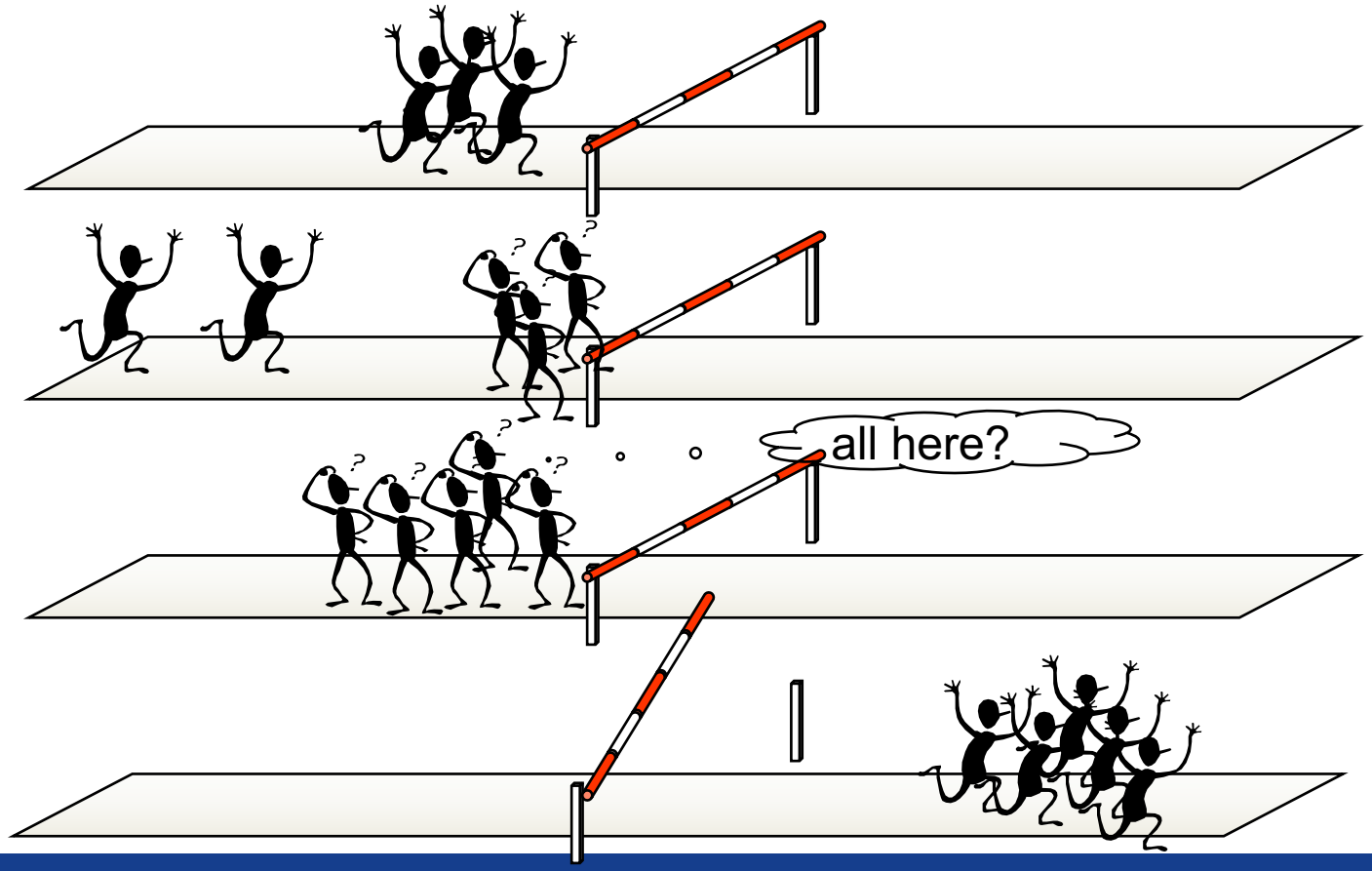
- The effective speed-up obtained by the parallelization depend by the amount of overhead we introduce making the algorithm parallel
- There are mainly two key sources of overhead:
  1. Time spent in inter-process interactions (**communication**)
  2. Time some process may spent being idle (**synchronization**)



Load Balancing

- Equally divide the work among the available resource: processors, memory, network bandwidth, I/O, ...
- This is usually a simple task for the problem decomposition model
- It is a difficult task for the functional decomposition model

Effect of Load Unbalancing



Minimizing Communication

- When possible reduce the communication events:
  - group lots of small communications into large one
  - eliminate synchronizations as much as possible. Each synchronization level off the performance to that of the slowest process

Overlap Communication and Computation

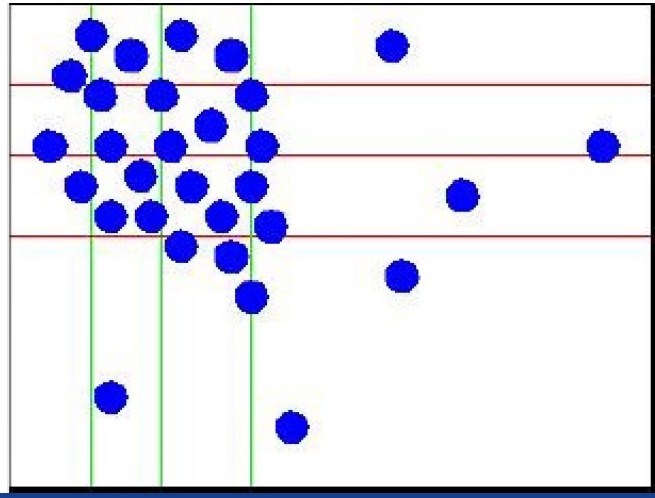
- When possible code your program in such a way that processes continue to do useful work while communicating
- This is usually a non trivial task and is afforded in the very last phase of parallelization
- If you succeed, you have done. Benefits are enormous

Granularity

- Granularity is determined by the decomposition level (number of task) on which we want divide the problem
- The degree to which task/data can be subdivided is limit to concurrency and parallel execution
- Parallelization has to become “topology aware”
  - coarse grain and fine grained parallelization has to be mapped to the topology to reduce memory and I/O contention
  - make your code modularized to enhance different levels of granularity and consequently to become more “platform adaptable”

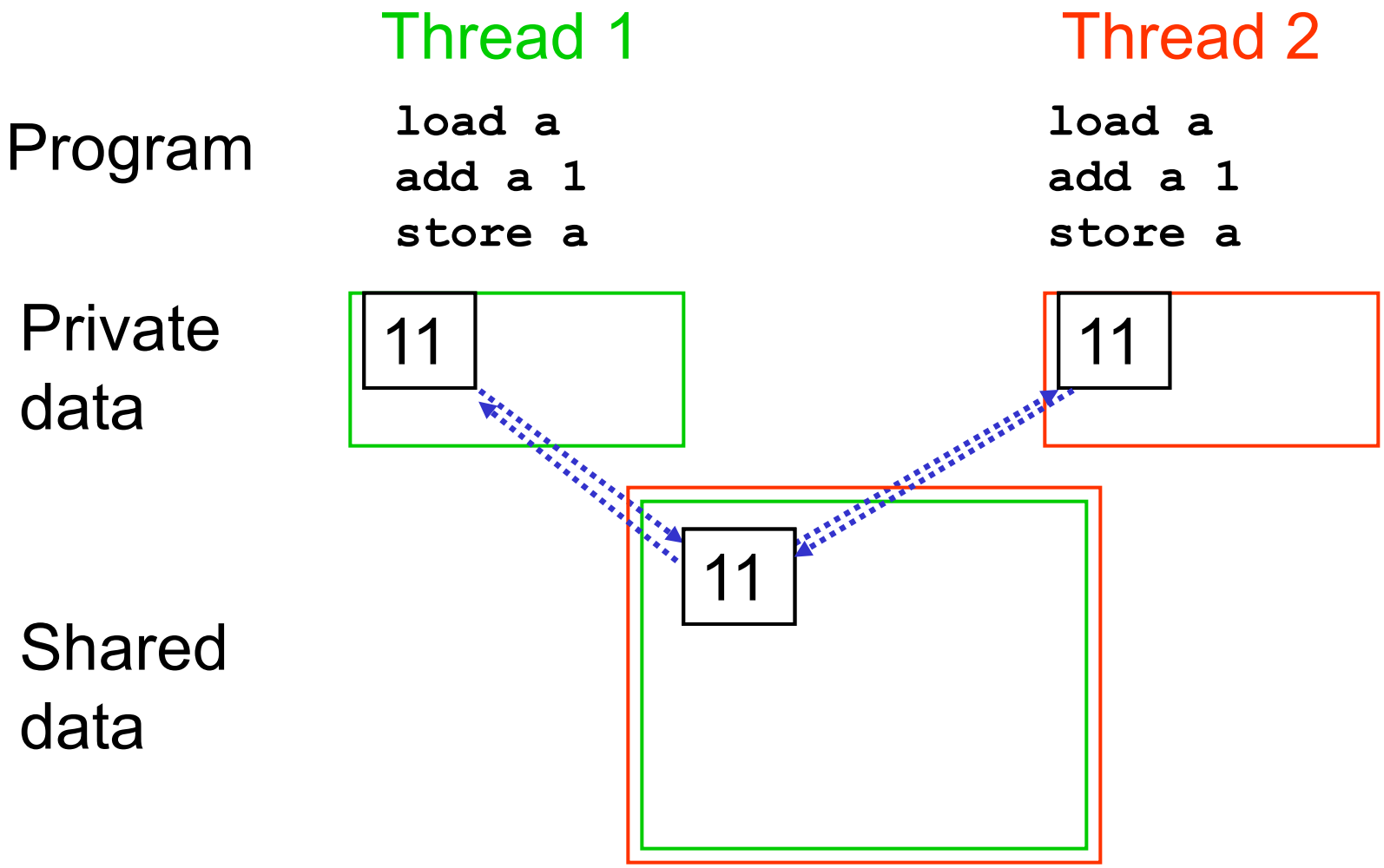
Limitations of Parallel Computing

- Fraction of serial code limits parallel speedup
- Degree to which tasks/data can be subdivided is limit to concurrency and parallel execution
- Load imbalance:
  - parallel tasks have a different amount of work
  - CPUs are partially idle
  - redistributing work helps but has limitations
  - communication and synchronization overhead



Shared Resources

- In parallel programming, developers must manage exclusive access to shared resources
- Resources are in different forms:
  - concurrent read/write (including parallel write) to shared memory locations
  - concurrent read/write (including parallel write) to shared devices
  - a message that must be send and received


Thread 1
Thread 2
Program
load a
add a 1
store a
load a
add a 1
store a
Private data
11
11
Shared data
11

Fundamental Tools of Parallel Programming

MPI Program Design

- Multiple and separate processes (can be local and remote) concurrently that are coordinated and exchange data through “messages”

  => a “share nothing” parallelization
- Best for coarse grained parallelization
- Distribute large data sets; replicate small data
- Minimize communication or overlap communication and computing for efficiency

  => Amdahl's law: speedup is limited by the fraction of serial code plus communication

Phases of an MPI Program

1. Startup
   - Parse arguments (mpirun may add some!)
   - Identify parallel environment and rank of process
   - Read and distribute all data
2. Execution
   - Proceed to subroutine with parallel work (can be same of different for all parallel tasks)
3. Cleanup

CAUTION: this sequence may be run only once

```
program bcast

  implicit none

  include "mpif.h"

  integer :: myrank, ncpus, imesg, ierr
  integer, parameter :: comm = MPI_COMM_WORLD

  call MPI_INIT(ierr)
  call MPI_COMM_RANK(comm, myrank, ierr)
  call MPI_COMM_SIZE(comm, ncpus, ierr)

  imesg = myrank
  print *, "Before Bcast operation I'm ", myrank, &
      " and my message content is ", imesg

  call MPI_BCAST(imesg, 1, MPI_INTEGER, 0, comm, ierr)

  print *, "After Bcast operation I'm ", myrank, &
      " and my message content is ", imesg

  call MPI_FINALIZE(ierr)

end program bcast
```

```
program bcast

  implicit none

  include "mpif.h"

  integer :: myrank, ncpus, imesg, ierr
  integer, parameter :: comm = MPI_COMM_WORLD
```

P_0

myrank = ??
ncpus = ??
imesg = ??
ierr = ??
comm = MPI_C...

P_1

myrank = ??
ncpus = ??
imesg = ??
ierr = ??
comm = MPI_C...

P_2

myrank = ??
ncpus = ??
imesg = ??
ierr = ??
comm = MPI_C...

P_3

myrank = ??
ncpus = ??
imesg = ??
ierr = ??
comm = MPI_C...

```
program bcast

  implicit none

  include "mpif.h"

  integer :: myrank, ncpus, imesg, ierr
  integer, parameter :: comm = MPI_COMM_WORLD

  call MPI_INIT(ierr)
```

P_0

myrank = ??
ncpus = ??
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

P_1

myrank = ??
ncpus = ??
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

P_2

myrank = ??
ncpus = ??
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

P_3

myrank = ??
ncpus = ??
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

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program bcast

  implicit none

  include "mpif.h"

  integer :: myrank, ncpus, imesg, ierr
  integer, parameter :: comm = MPI_COMM_WORLD

  call MPI_INIT(ierr)

  call MPI_COMM_SIZE(comm, ncpus, ierr)

  call MPI_COMM_RANK(comm, myrank, ierr)
```

P_0

myrank = ??
ncpus = 4
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

P_1

myrank = ??
ncpus = 4
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

P_2

myrank = ??
ncpus = 4
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

P_3

myrank = ??
ncpus = 4
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

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program bcast

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  integer :: myrank, ncpus, imesg, ierr
  integer, parameter :: comm = MPI_COMM_WORLD

  call MPI_INIT(ierr)

  call MPI_COMM_SIZE(comm, ncpus, ierr)

  call MPI_COMM_RANK(comm, myrank, ierr)
```

P_0

myrank = 0
ncpus = 4
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

P_1

myrank = 1
ncpus = 4
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

P_2

myrank = 2
ncpus = 4
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

P_3

myrank = 3
ncpus = 4
imesg = ??
ierr = MPI_SUC...
comm = MPI_C...

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program bcast

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  integer :: myrank, ncpus, imesg, ierr
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P_0

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imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_1

myrank = 1
ncpus = 4
imesg = 1
ierr = MPI_SUC...
comm = MPI_C...

P_2

myrank = 2
ncpus = 4
imesg = 2
ierr = MPI_SUC...
comm = MPI_C...

P_3

myrank = 3
ncpus = 4
imesg = 3
ierr = MPI_SUC...
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program bcast

  implicit none

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  integer :: myrank, ncpus, imesg, ierr
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  imesg = myrank
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  call MPI_BCAST(imesg, 1, MPI_INTEGER, 0, comm, ierr)
```

P_0

myrank = 0
ncpus = 4
imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_1

myrank = 1
ncpus = 4
imesg = 1
ierr = MPI_SUC...
comm = MPI_C...

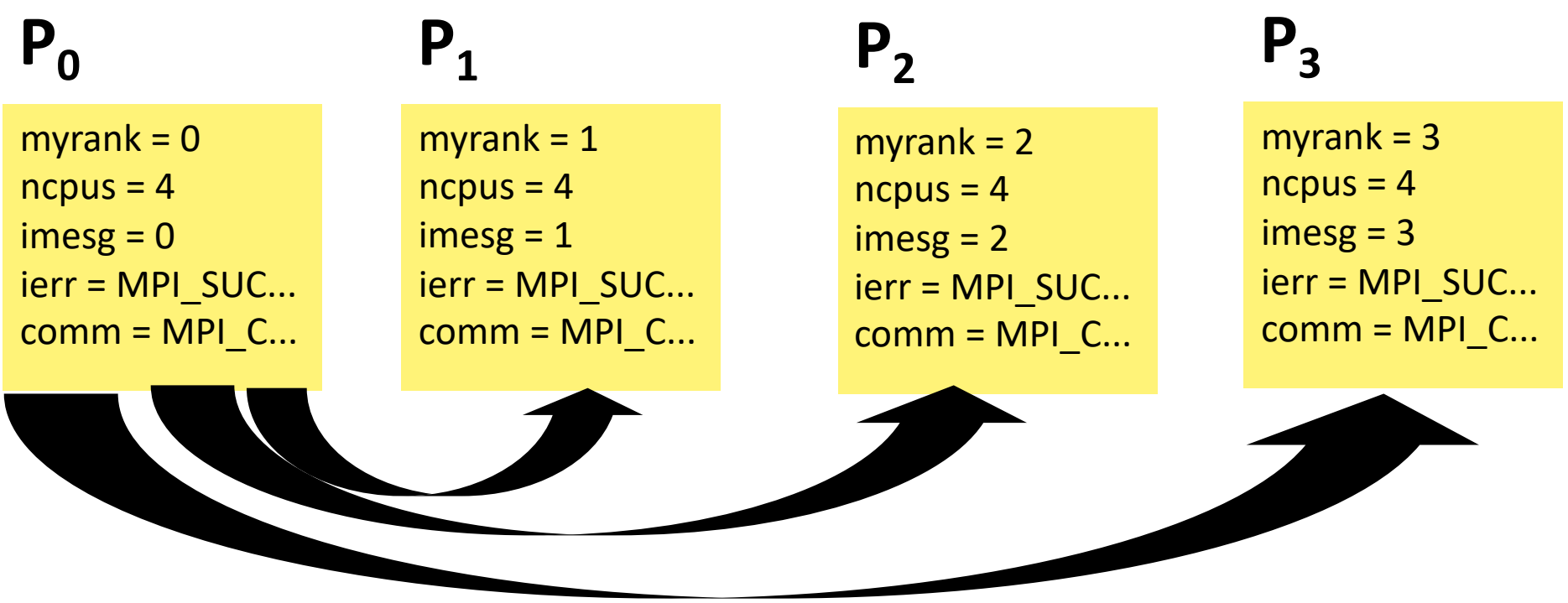
P_2

myrank = 2
ncpus = 4
imesg = 2
ierr = MPI_SUC...
comm = MPI_C...

P_3

myrank = 3
ncpus = 4
imesg = 3
ierr = MPI_SUC...
comm = MPI_C...

call MPI_BCAST(imesg, 1, MPI_INTEGER, 0, comm, ierr)



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P_0

myrank = 0
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imesg = 0
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comm = MPI_C...

P_1

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imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_2

myrank = 2
ncpus = 4
imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_3

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ncpus = 4
imesg = 0
ierr = MPI_SUC...
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P_1

myrank = 1
ncpus = 4
imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_2

myrank = 2
ncpus = 4
imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_3

myrank = 3
ncpus = 4
imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

```
program bcast

  implicit none

  include "mpif.h"

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P_0

myrank = 0
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comm = MPI_C...

P_1

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imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_2

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ncpus = 4
imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_3

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ncpus = 4
imesg = 0
ierr = MPI_SUC...
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      " and my message content is ", imesg

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  call MPI_FINALIZE(ierr)

end program bcast
```

P_0

myrank = 0
ncpus = 4
imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_1

myrank = 1
ncpus = 4
imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

P_2

myrank = 2
ncpus = 4
imesg = 0
ierr = MPI_SUCC
comm = MPI_C...

P_3

myrank = 3
ncpus = 4
imesg = 0
ierr = MPI_SUC...
comm = MPI_C...

Programming Parallel Paradigms

- Are the tools we use to express the parallelism for on a given architecture (see also SPMD, SIMD, etc...)
- They differ in how programmers can manage and define key features like:
  - parallel regions
  - concurrency
  - process communication
  - synchronism

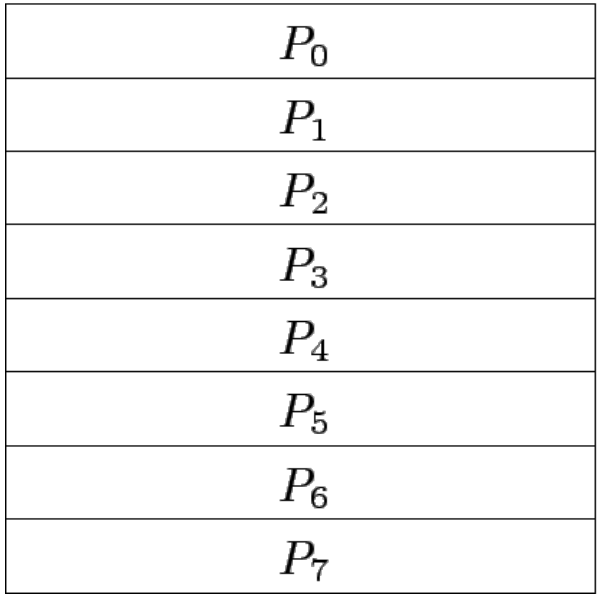
Parallelism - 101

- there are two main reasons to write a parallel program:
  - access to larger amount of memory (aggregated, going bigger)
  - reduce time to solution (going faster)

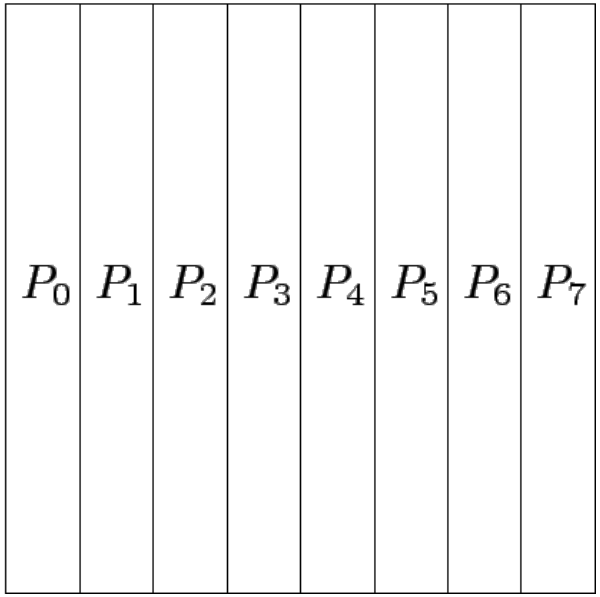
Static Data Partitioning

The simplest data decomposition schemes for dense matrices are 1-D block distribution schemes.

row-wise distribution



column-wise distribution

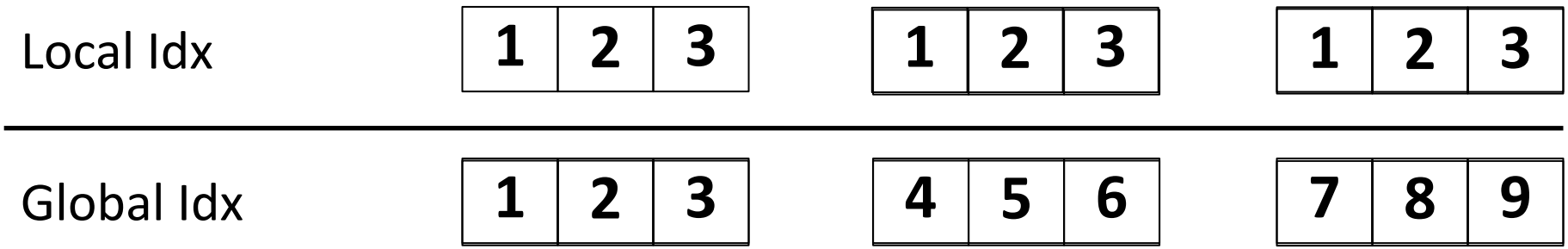


Distributed Data Vs Replicated Data

- Replicated data distribution is useful if it helps to reduce the communication among process at the cost of bounding scalability
- Distributed data is the ideal data distribution but not always applicable for all data-sets
- Usually complex application are a mix of those techniques => distribute large data sets; replicate small data

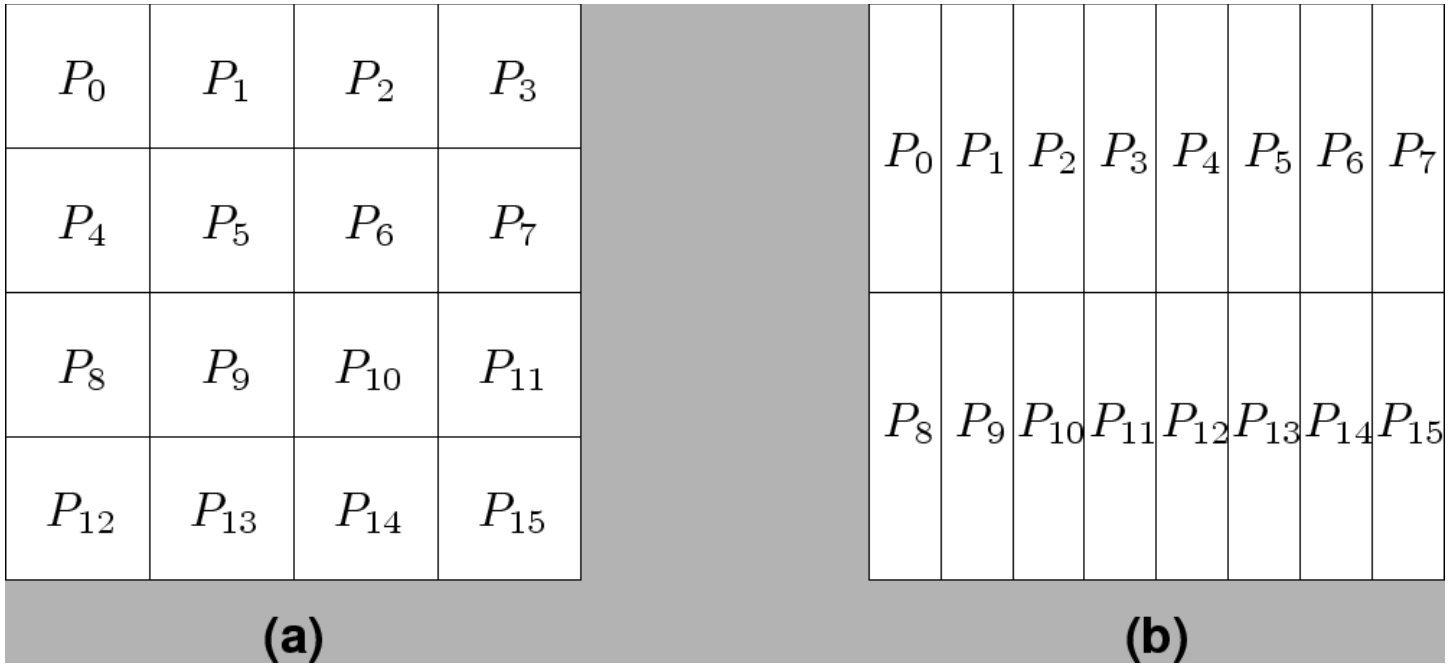
Global Vs Local Indexes

- In sequential code you always refer to global indexes
- With distributed data you must handle the distinction between global and local indexes (and possibly implementing utilities for transparent conversion)



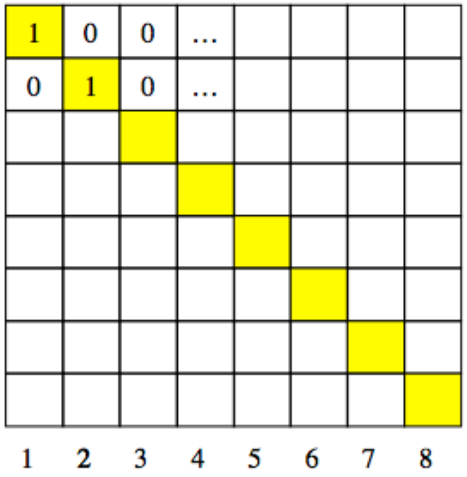
Block Array Distribution Schemes

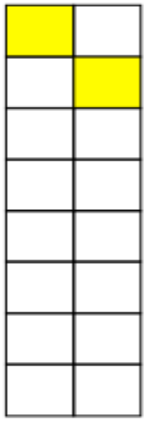
Block distribution schemes can be generalized to higher dimensions as well.

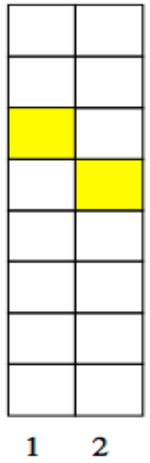


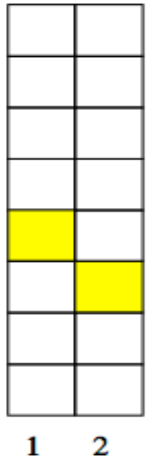
Degree to which tasks/data can be subdivided is limit to concurrency and parallel execution!!

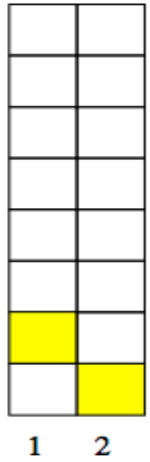
Collaterals to Domain Decomposition /1







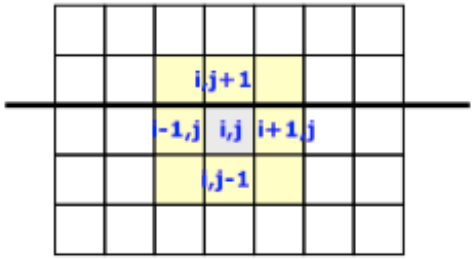


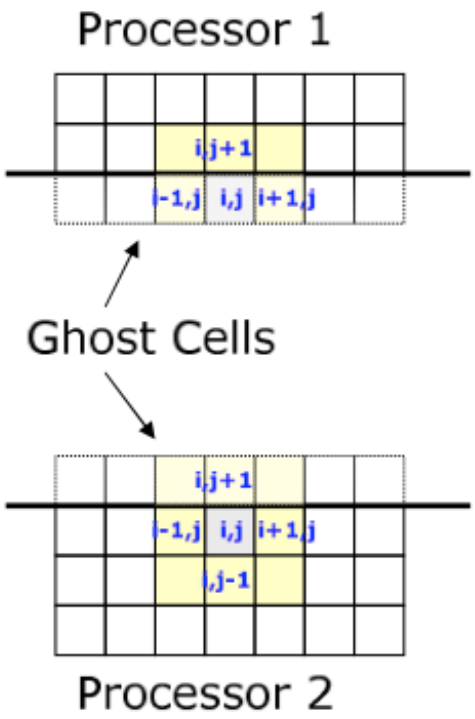


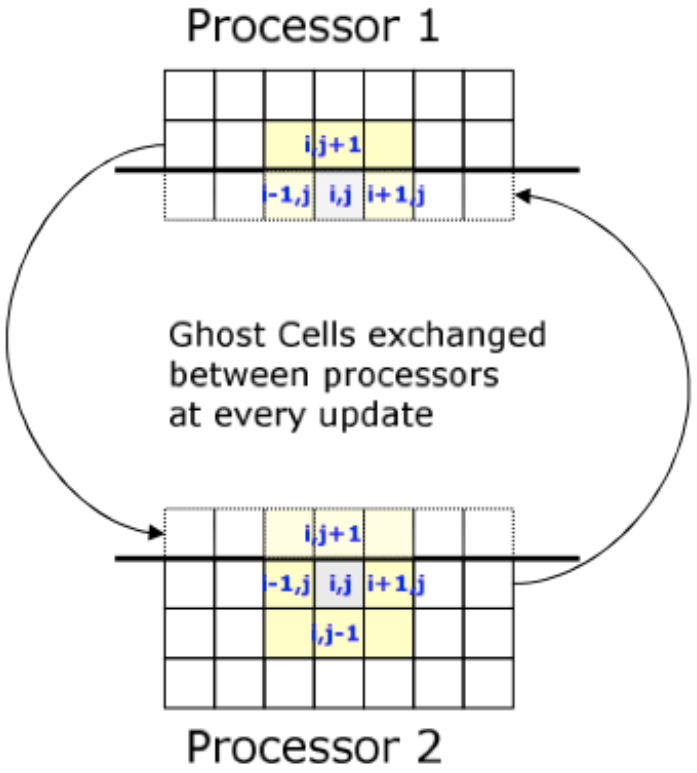
Are all the domain's dimensions always multiple of the number of tasks/processes we are willing to use?

Again on Domain Decomposition

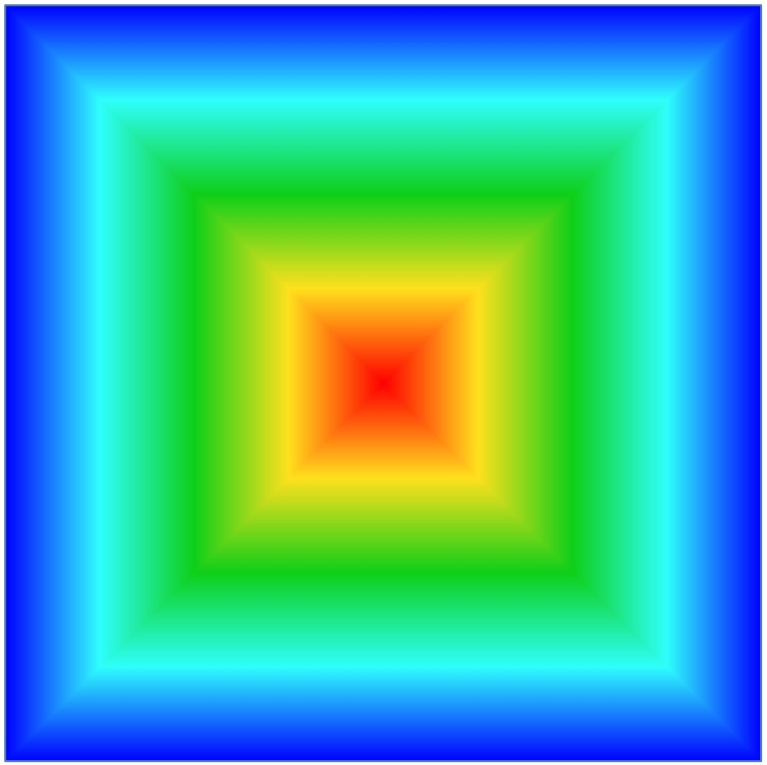
sub-domain boundaries



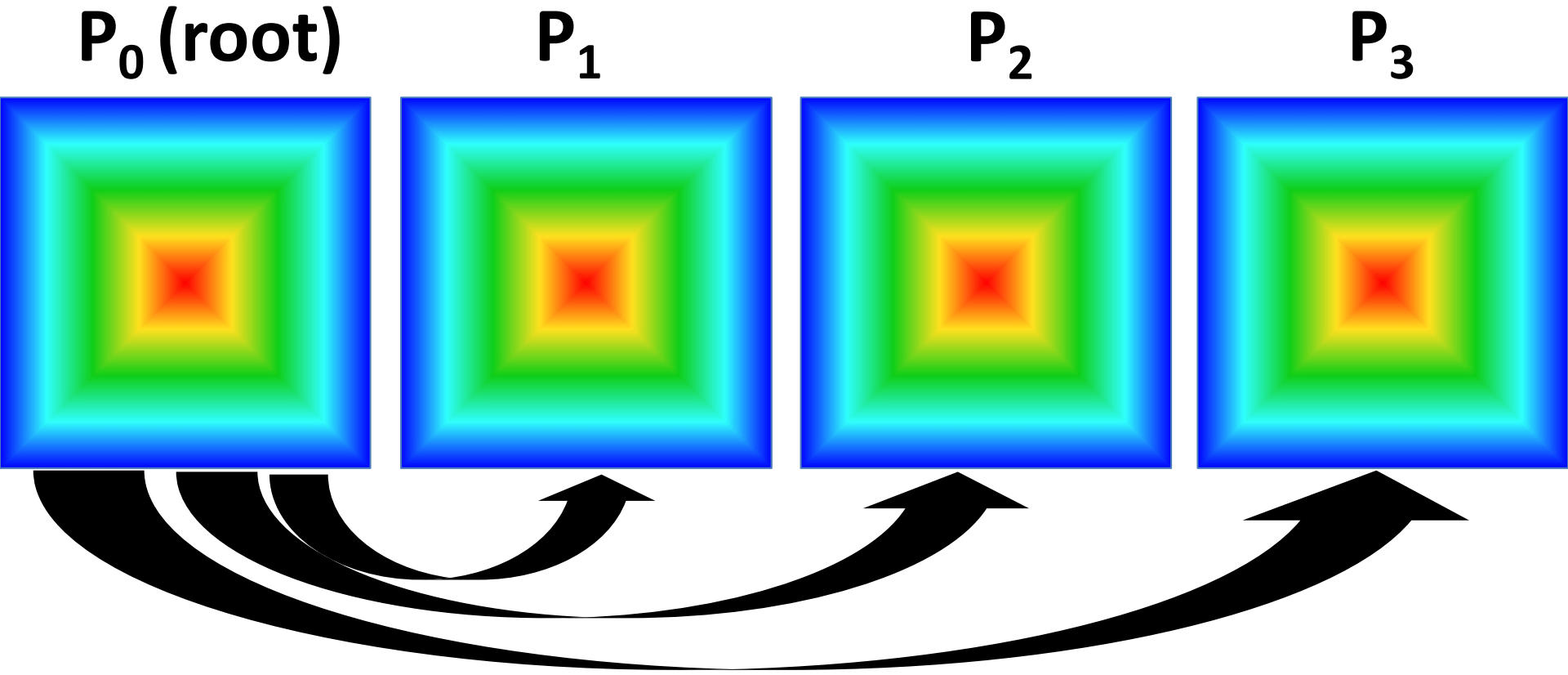


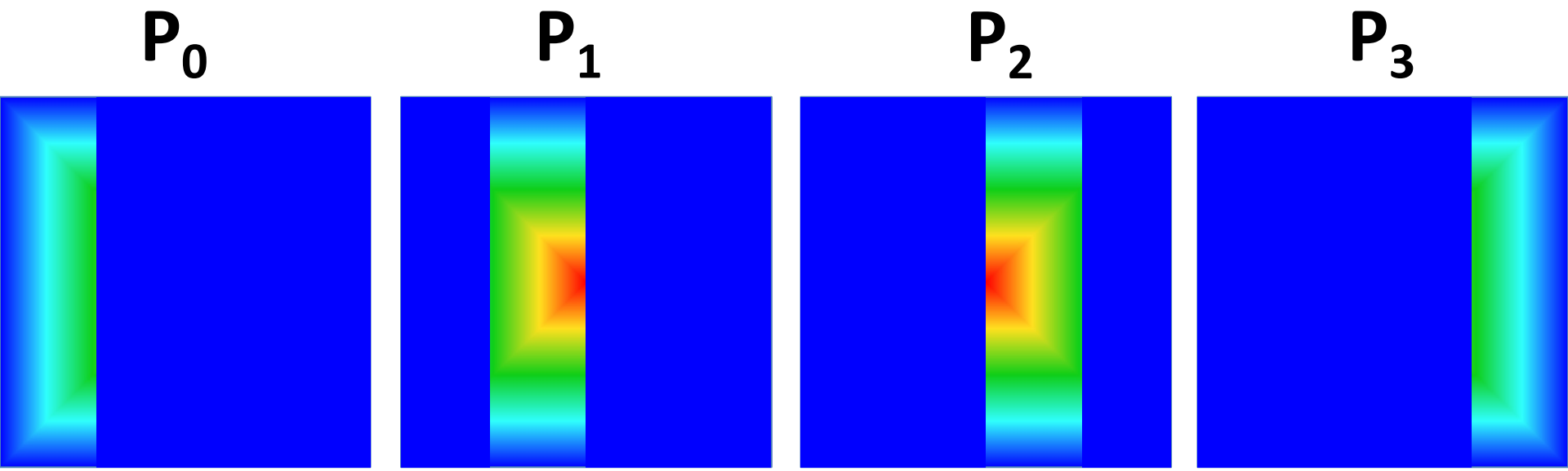


P_0



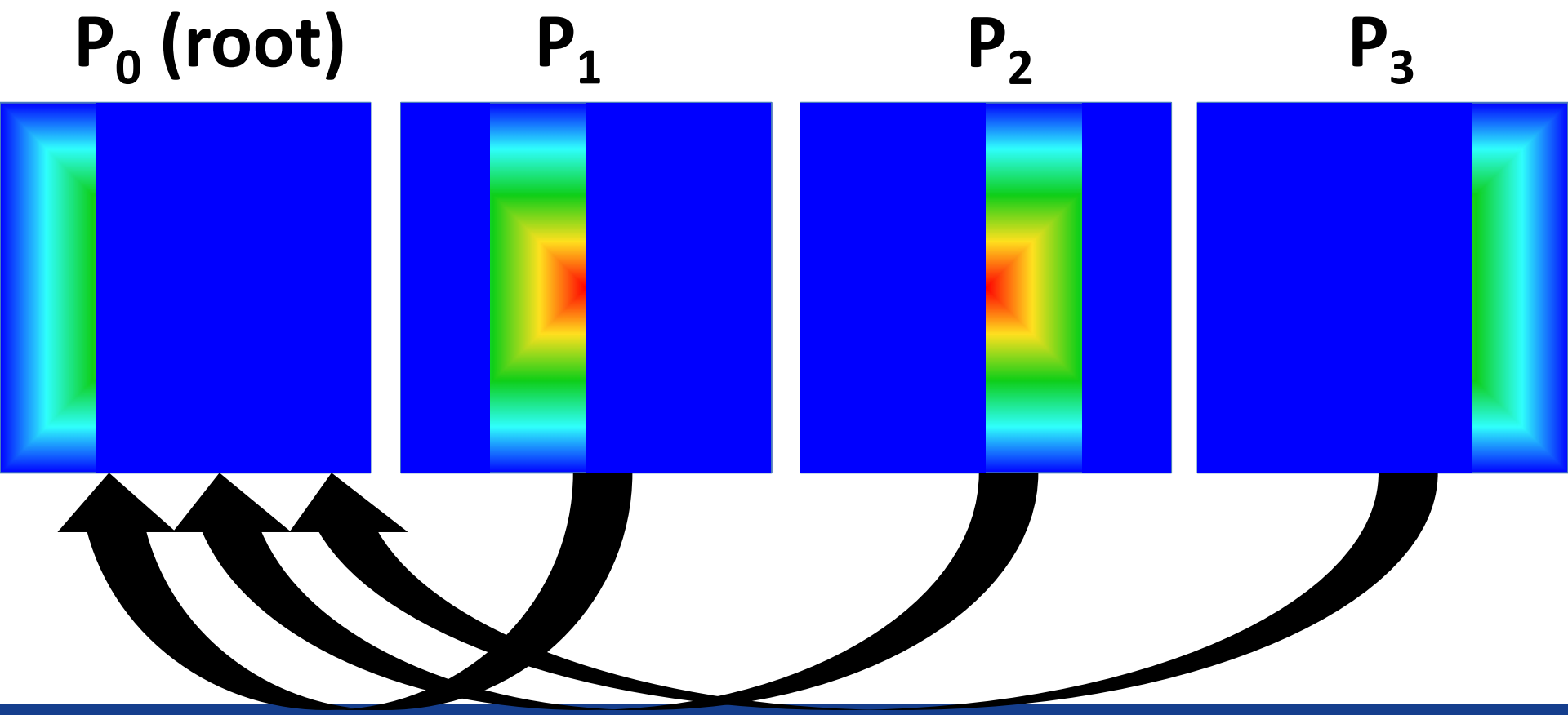
call MPI_BCAST(...)





call evolve(dtfact)

call MPI_Gather(…, …, …)

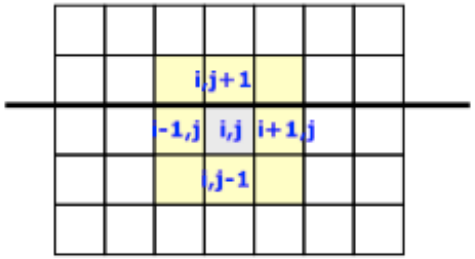


Replicated data

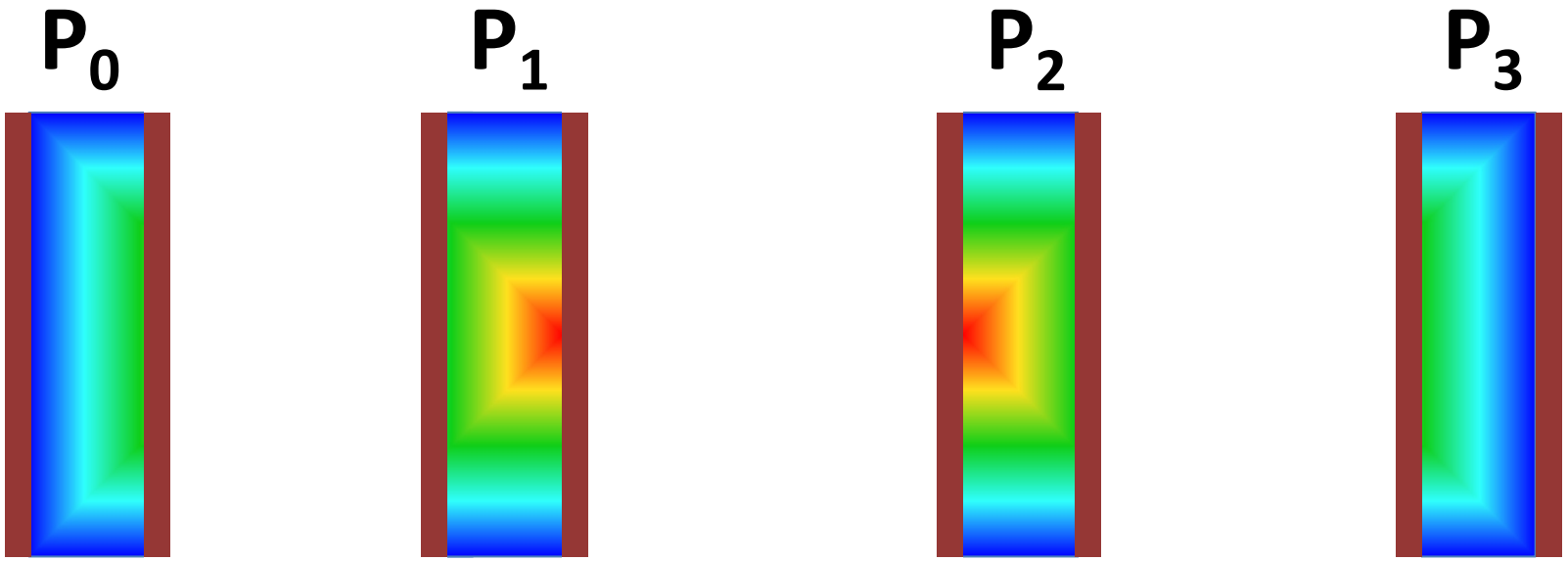
- Compute domain (and workload) distribution among processes
- Master-slaves: P_0 drives all processes
- Large amount of data communication
  - at each step P_0 distribute data to all processes and collect the contribution of each process
- Problem size scaling limited in memory capacity

Collaterals to Domain Decomposition /2

sub-domain boundaries

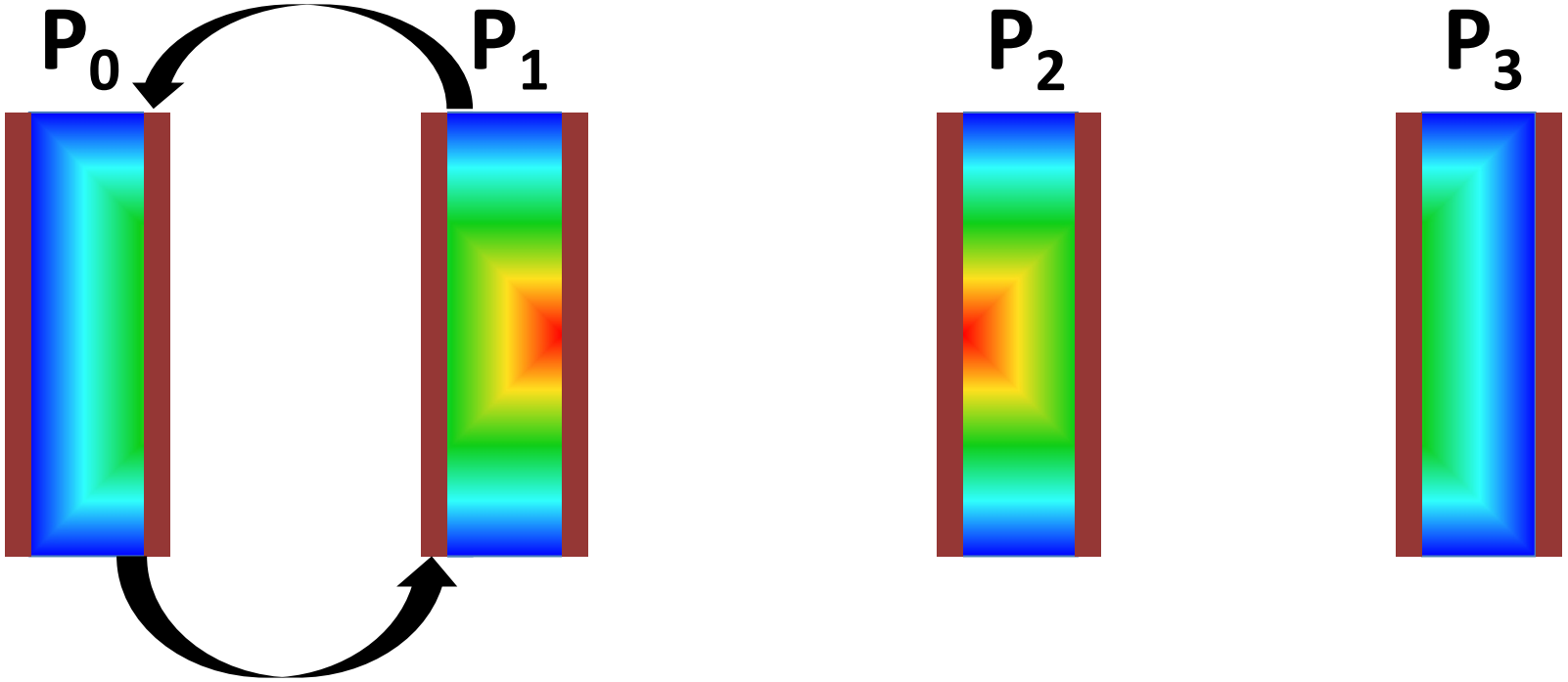


The Transport Code - Parallel Version

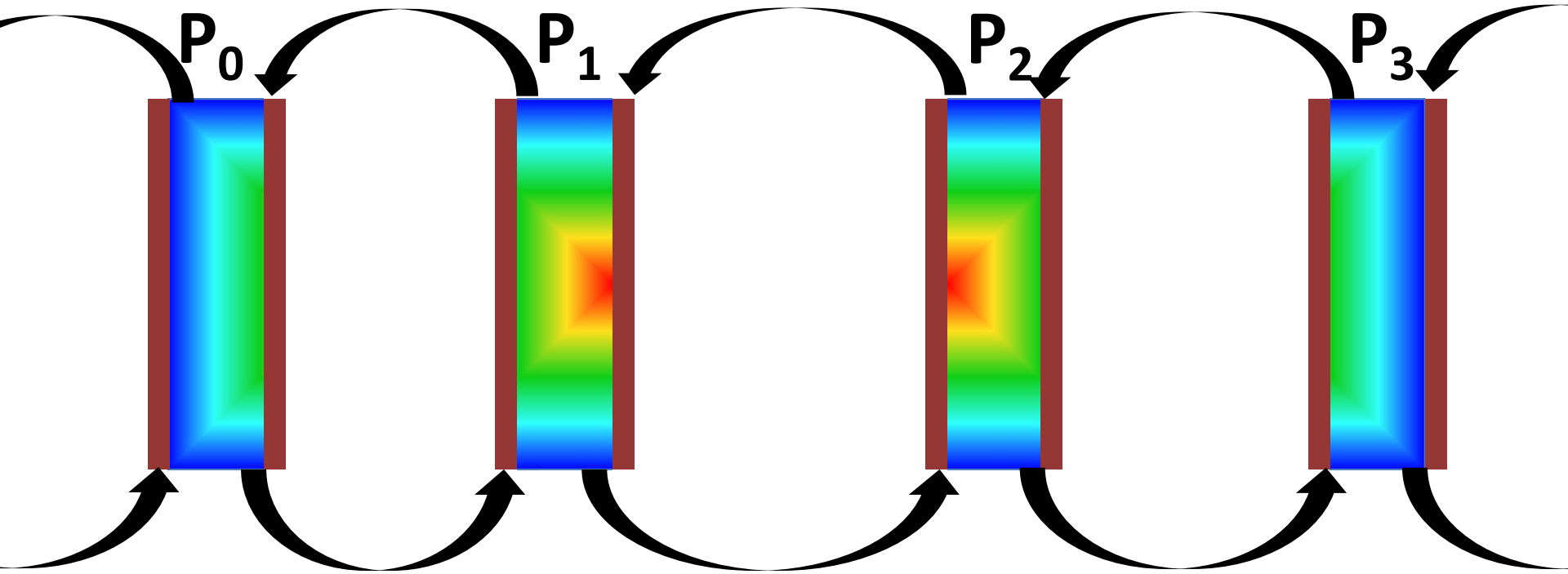


call evolve(dtfact)

Data exchange among processes



proc_down = mod(proc_me - 1 + nprocs , nprocs)

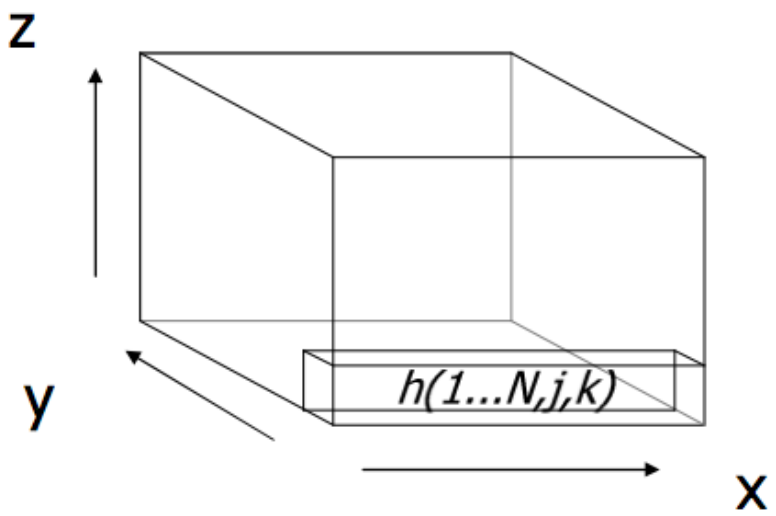


proc_up = mod(proc_me + 1 , nprocs)

Distributed Data

- Global and Local Indexes
- Ghost Cells Exchange Between Processes
  - Compute Neighbor Processes
- Parallel Output

Multidimensional FFT



$$f(x,y,z) = \frac{1}{N_z N_y N_x} \sum_{z=0}^{N_z-1} \Big(\sum_{y=0}^{N_y-1} \Big(\underbrace{\sum_{x=0}^{N_x-1} F(u,v,w)\, e^{-2\pi i \frac{xu}{N_x}}}_{\text{DFT long x-dimension}} \Big) e^{-2\pi i \frac{yv}{N_y}} \Big)_{\text{DFT long y-dimension}} e^{-2\pi i \frac{zw}{N_z}}$$

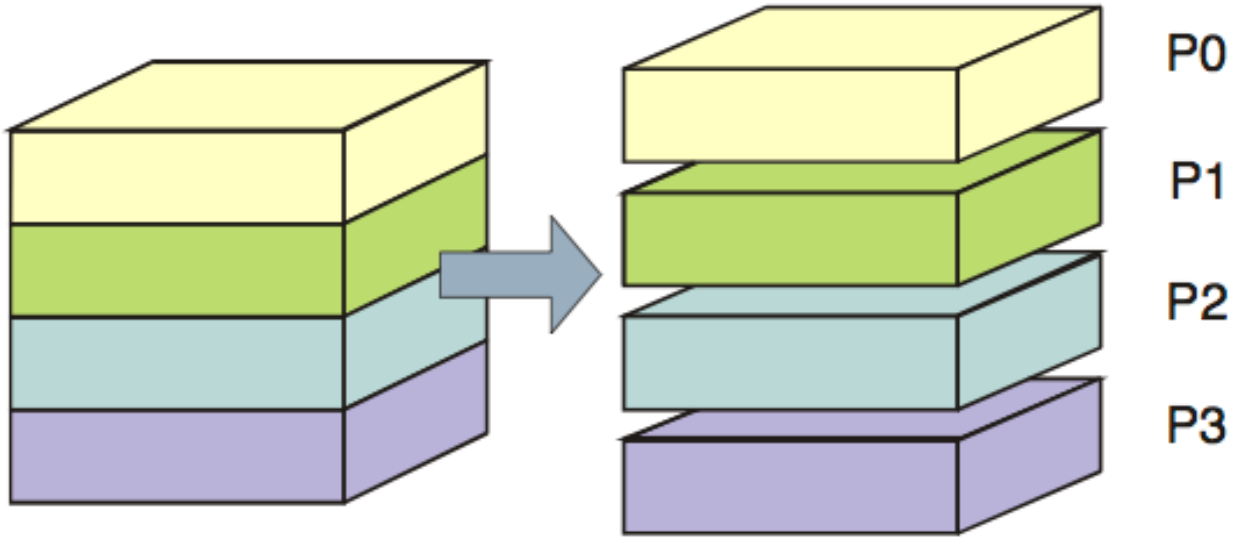
DFT long z-dimension

1) For any value of ***j*** and ***k*** transform the column (1…N, j, k)

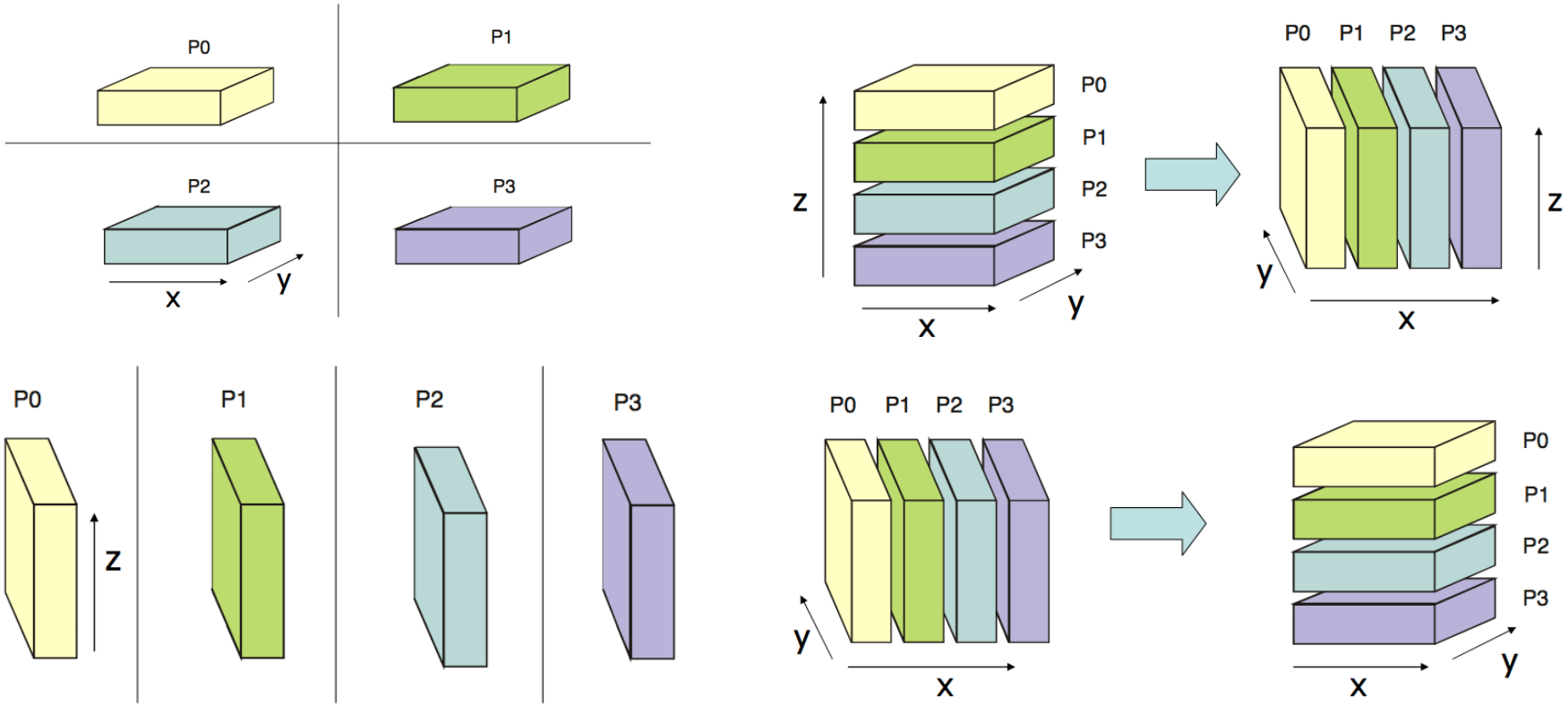
2) For any value of ***i*** and ***k*** transform the column (i, 1…N, k)

3) For any value of ***i*** and ***j*** transform the column (i, j, 1…N)

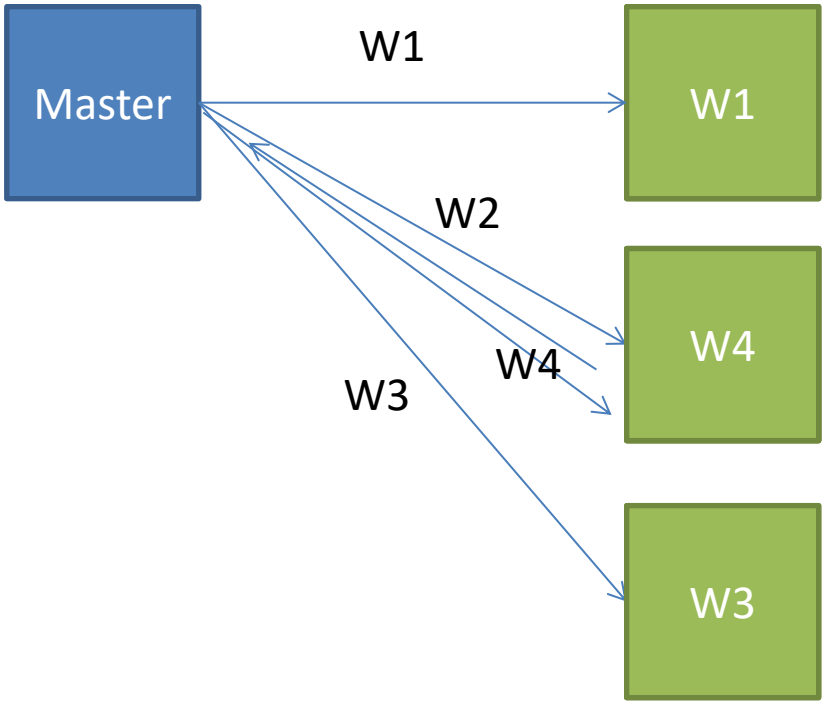
Parallel 3DFFT / 1



Parallel 3DFFT / 2



Master/Slave



Task Farming

- Many independent programs (tasks) running at once
  - each task can be serial or parallel
  - "independent" means they don't communicate directly
  - Processes possibly driven by the mpirun framework

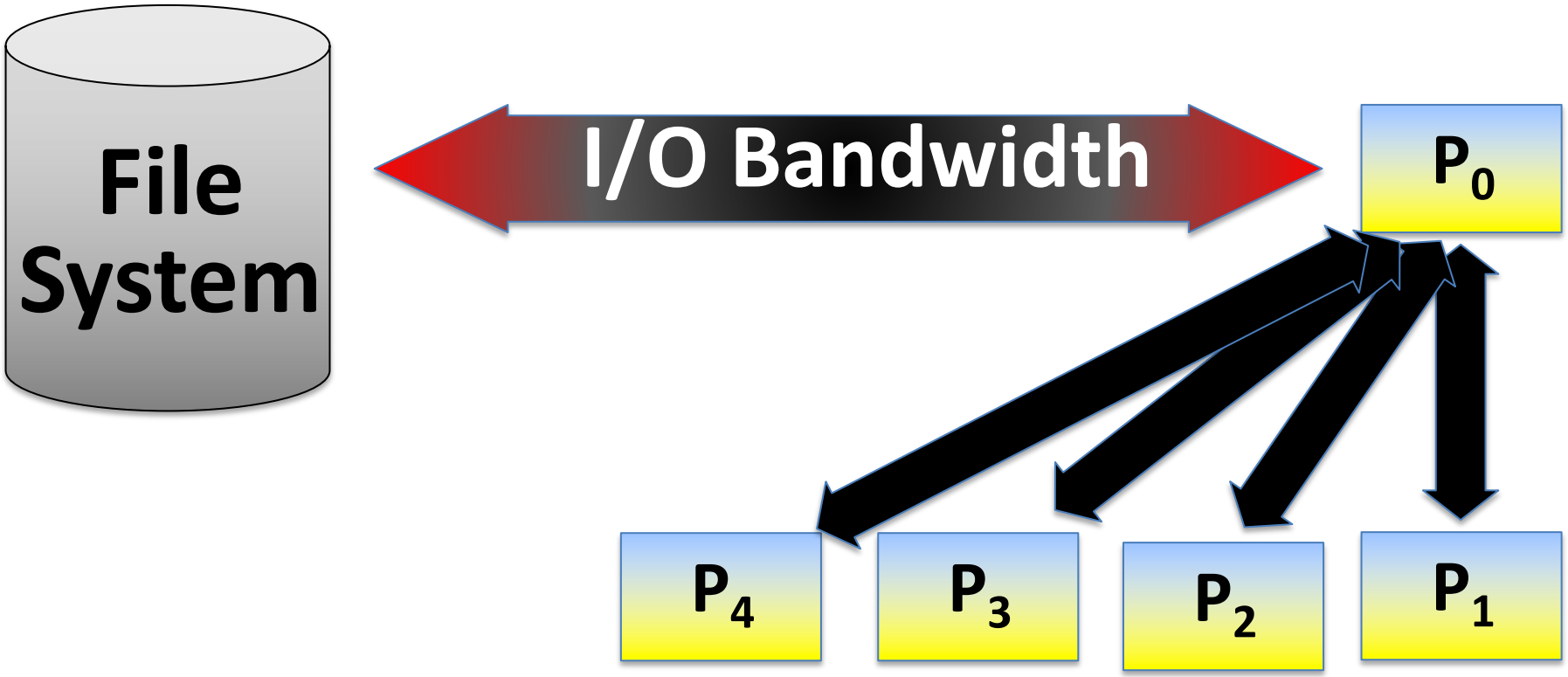
```
[igirotto@localhost]$ more my_shell_wrapper.sh
#!/bin/bash
#example for the OpenMPI implementation
./prog.x --input input_${OMPI_COMM_WORLD_RANK}.dat

[igirotto@localhost]$ mpirun -np 400 ./my_shell_wrapper.sh
```

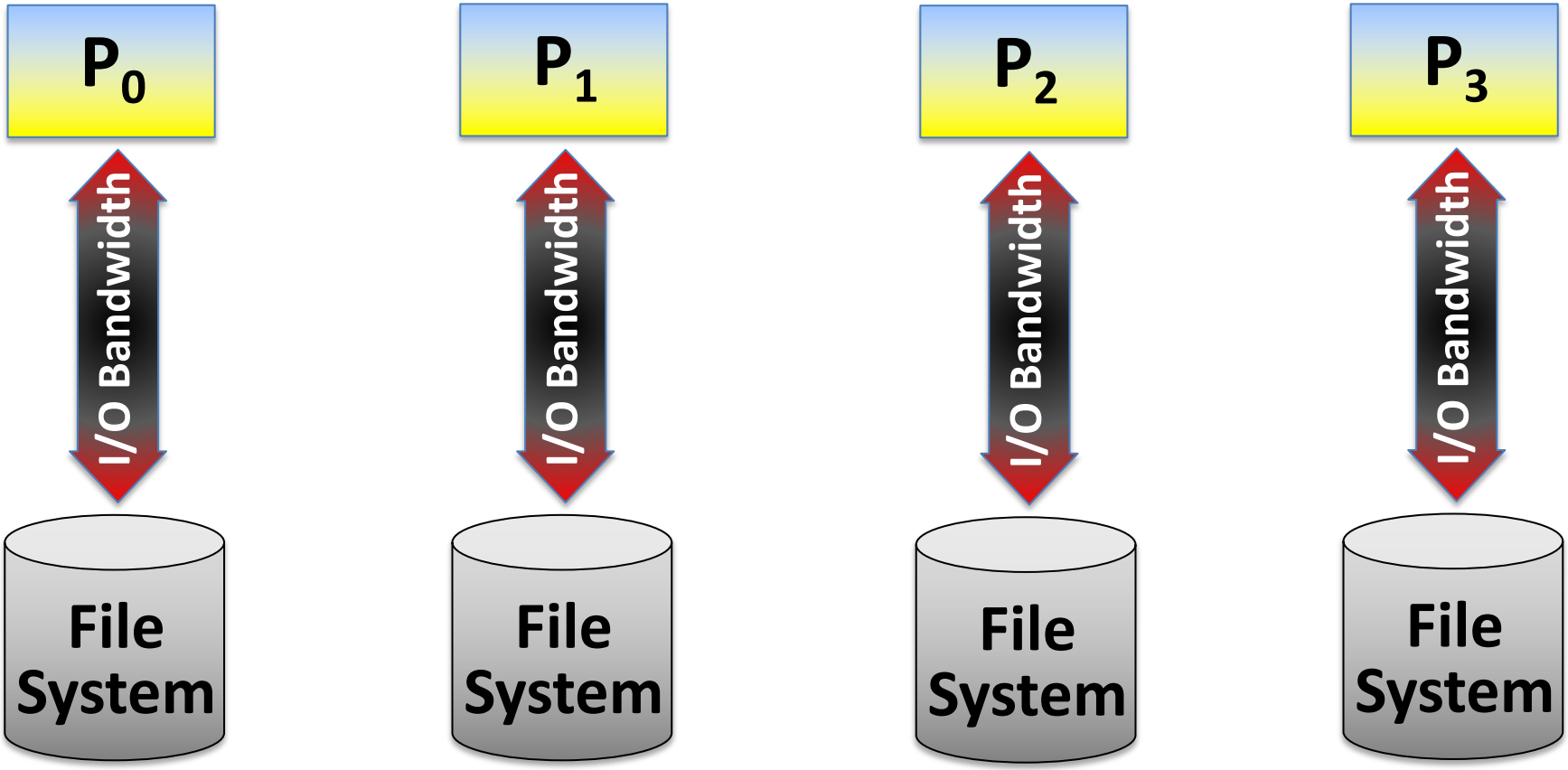
Easy Parallel Computing

- Farming, embarrassingly parallel
  - Executing multiple instances on the same program with different inputs/initial cond.
  - Reading large binary files by splitting the workload among processes
  - Searching elements on large data-sets
  - Other parallel execution of embarrassingly parallel problem (no communication among tasks)
- Ensemble simulations (weather forecast)
- Parameter space (find the best wing shape)

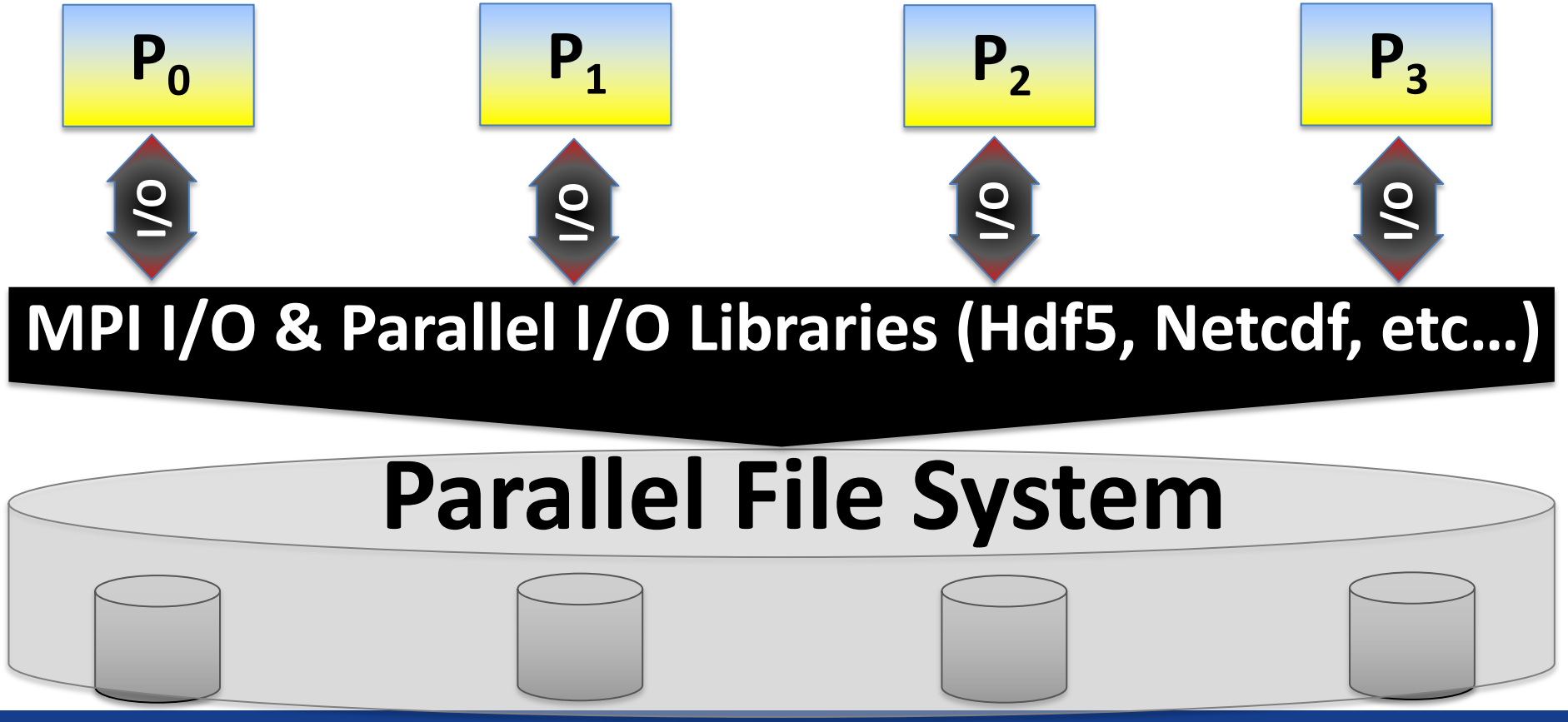
Parallel I/O



Parallel I/O


P0
P1
P2
P3
I/O Bandwidth
I/O Bandwidth
I/O Bandwidth
I/O Bandwidth
File System
File System
File System
File System

Parallel I/O


P0
P1
P2
P3
I/O
I/O
I/O
I/O
MPI I/O & Parallel I/O Libraries (Hdf5, Netcdf, etc…)
Parallel File System

Make Use Freely Available Parallel Libraries

- Scalable Parallel Random Number Generators Library (SPRNG)
- Parallel Linear Algebra (ScaLAPACK)
- Parallel Library for Solution of Finite Elements (dealii)
- Parallel Library for FFT (FFTW)
- Parallel Linear Solver for Sparce Matrices (PETSc)

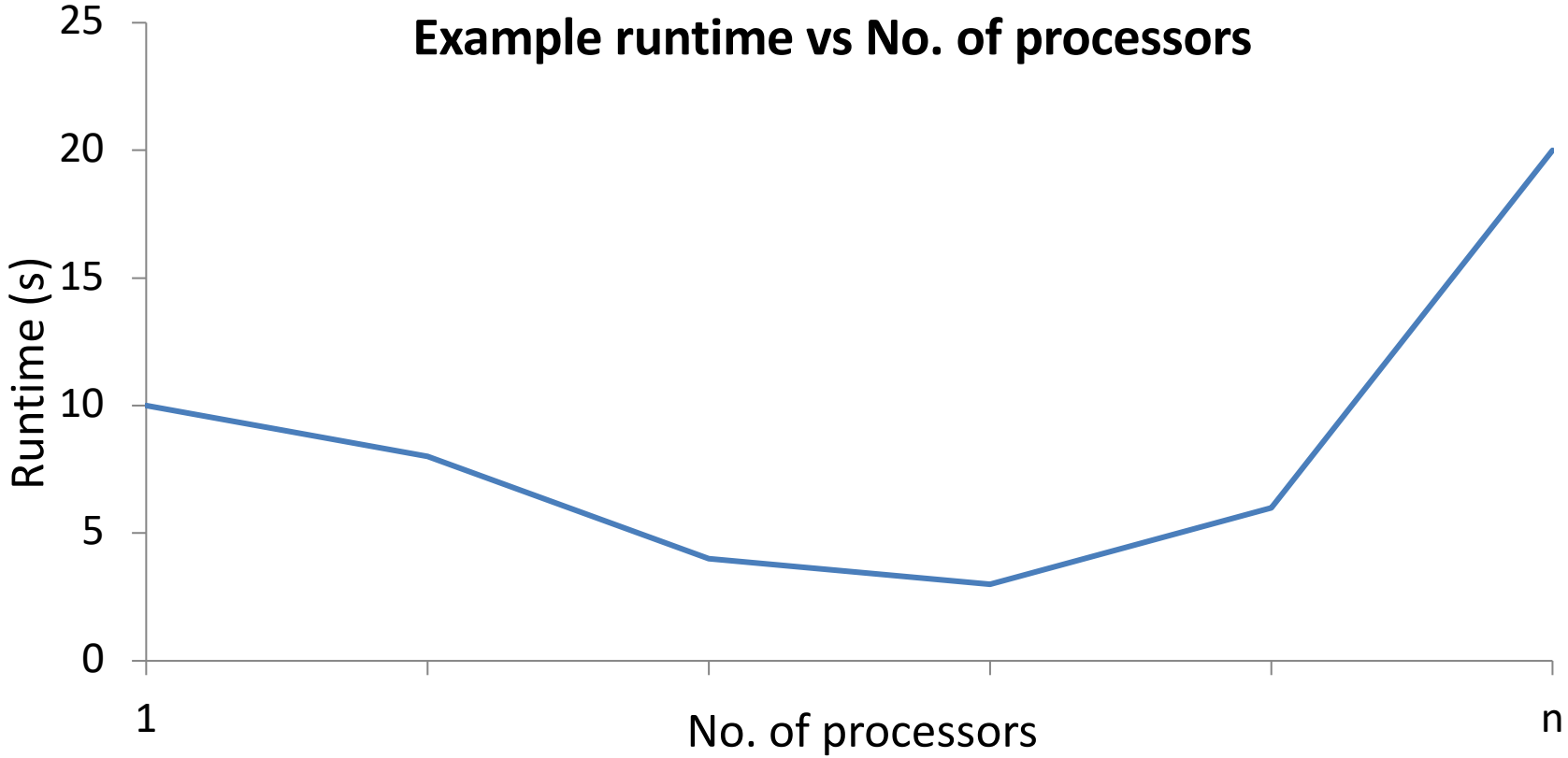
Measure of Parallelism

- evaluation of the performance improvement is not always straightforward, specially considering at the increasing the problem complexity

Scaling

- *Scaling* is how the performance of a parallel application changes as the number of processors is increased
- There are two different types of scaling:
  - *Strong Scaling* – total problem size stays the same as the number of processors increases
  - *Weak Scaling* – the problem size increases at the same rate as the number of processors, keeping the amount of work per processor the same
- Strong scaling is generally more useful and more difficult to achieve than weak scaling

Strong scaling

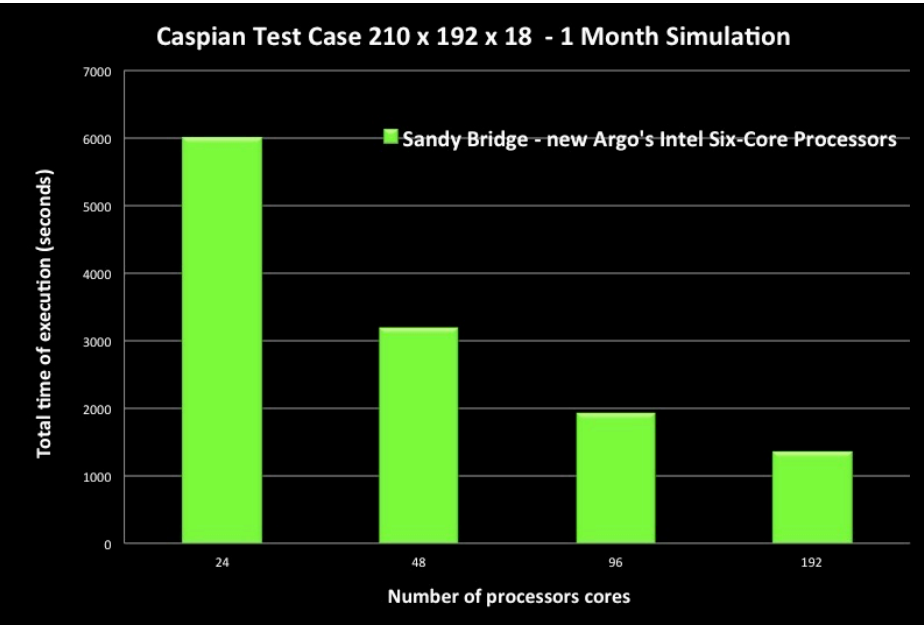


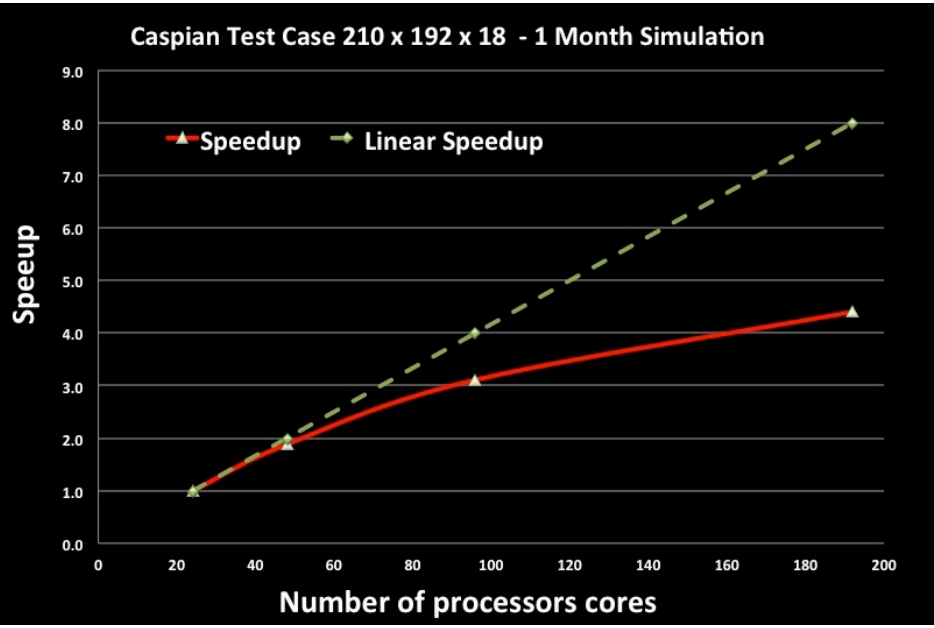
How do we evaluate the improvement?

- We want estimate the amount of the introduced overhead => $T_o = n_{pes} T_P - T_S$
- But to quantify the improvement we use the term **Speedup**:

$$S_P = \frac{T_S}{T_P}$$

Speedup


Caspian Test Case 210 x 192 x 18 - 1 Month Simulation
Sandy Bridge - new Argo's Intel Six-Core Processors
Total time of execution (seconds)
7000
6000
5000
4000
3000
2000
1000
0
24
48
96
192
Number of processors cores

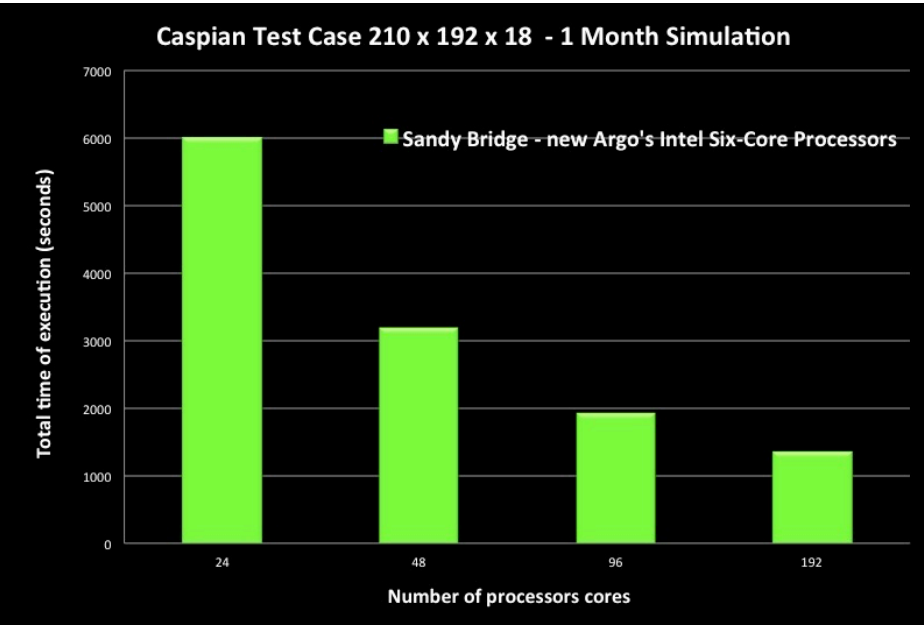

Caspian Test Case 210 x 192 x 18 - 1 Month Simulation
Speedup
Linear Speedup
Speeup
9.0
8.0
7.0
6.0
5.0
4.0
3.0
2.0
1.0
0.0
0
20
40
60
80
100
120
140
160
180
200
Number of processors cores

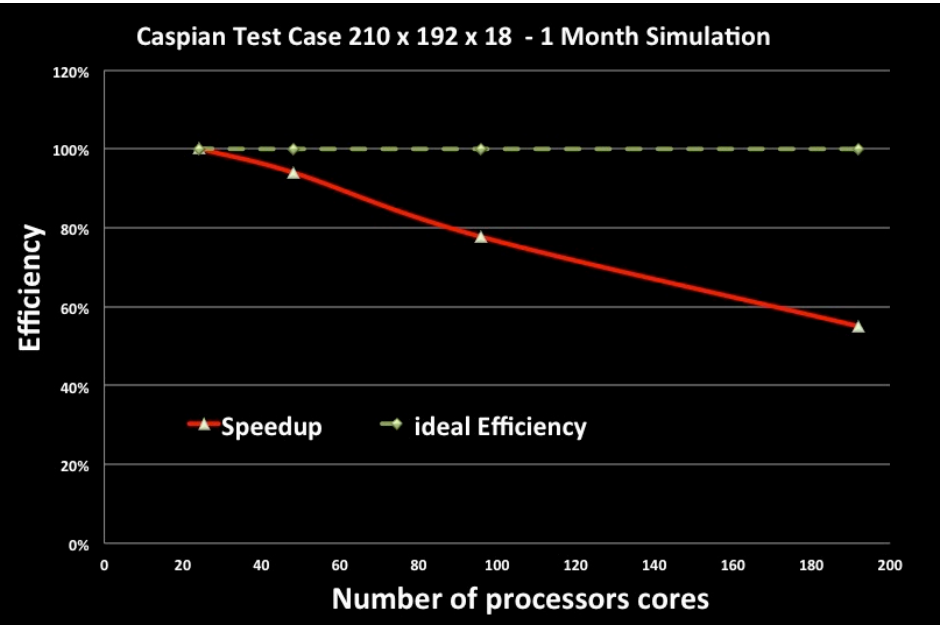
Efficiency

- Only embarrassing parallel algorithm can obtain an ideal Speedup
- The **Efficiency** is a measure of the fraction of time for which a processing element is usefully employed:

$$E_P = \frac{S_P}{p}$$

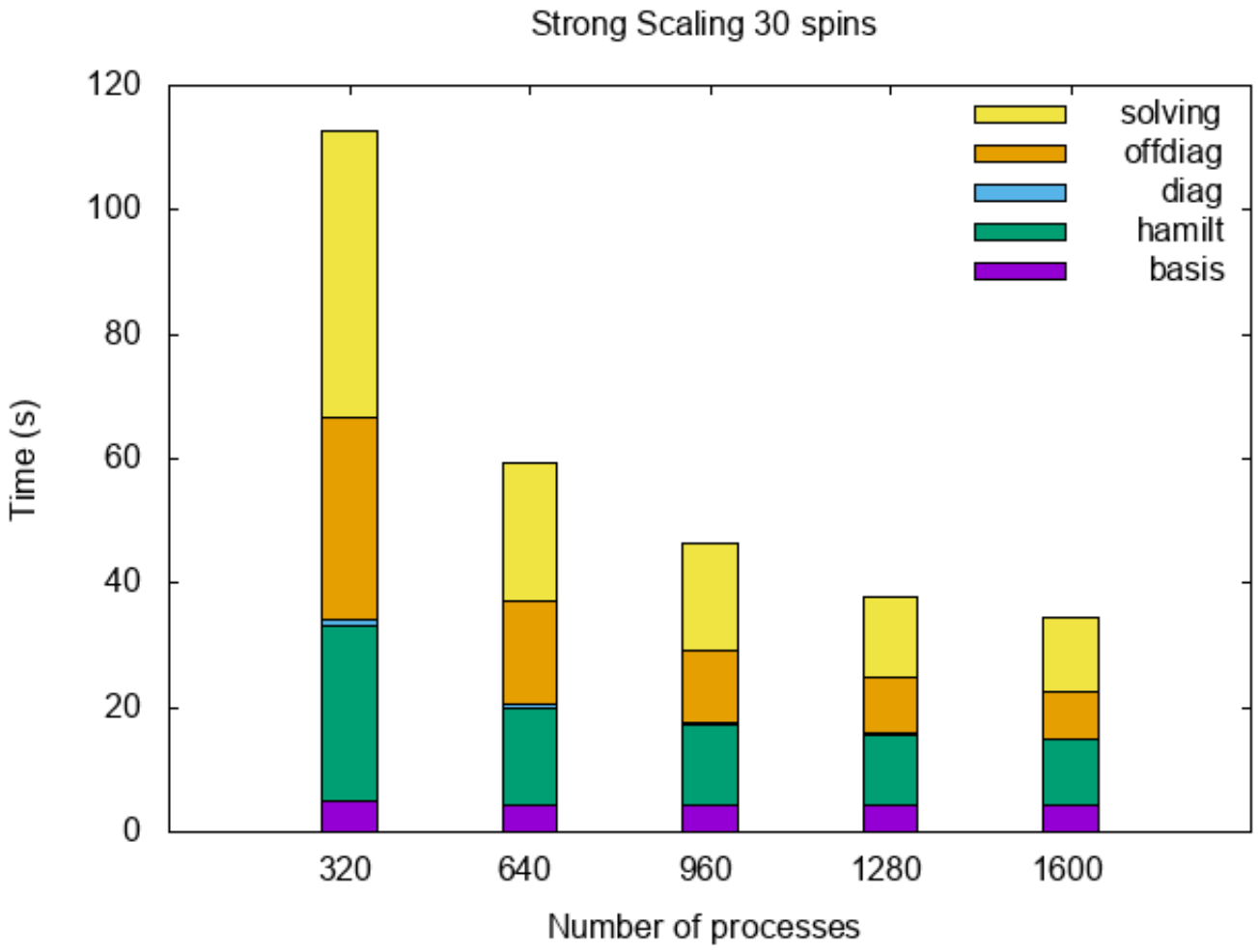
Efficiency


Caspian Test Case 210 x 192 x 18 - 1 Month Simulation
Total time of execution (seconds)
7000
6000
5000
4000
3000
2000
1000
0
Sandy Bridge - new Argo's Intel Six-Core Processors
24
48
96
192
Number of processors cores


Caspian Test Case 210 x 192 x 18 - 1 Month Simulation
Efficiency
120%
100%
80%
60%
40%
20%
0%
Speedup
ideal Efficiency
0
20
40
60
80
100
120
140
160
180
200
Number of processors cores

Scalability Limits

- Amdahl's law (there is a serial part which is not effected)
- Parallelism introduces overhead due to communication, synchronization, additional operations (I/O, memory copies, reordering, etc...)
- Sometimes the limit of scalability can be somehow predicted:
  - is there enough work for any process?
- When considering complex problem not all part of the problem scale eqaully


Strong Scaling 30 spins
Time (s)
0
20
40
60
80
100
120
320
640
960
1280
1600
Number of processes
solving
offdiag
diag
hamilt
basis

Timing

- Time measurement becomes a relevant task:
- MPI_WTIME()
- Clock
- Seconds()

```
     convergence NOT achieved after   5 iterations: stopping

     Writing output data file c8_atm213_k111.save

     init_run     :     93.79s CPU     93.79s WALL (       1 calls)
     electrons    :    961.37s CPU    961.37s WALL (       1 calls)

     Called by init_run:
     wfcinit      :     69.37s CPU     69.37s WALL (       1 calls)
     potinit      :      4.76s CPU      4.76s WALL (       1 calls)

     Called by electrons:
     c_bands      :    883.32s CPU    883.32s WALL (       5 calls)
     sum_band     :     40.30s CPU     40.30s WALL (       5 calls)
     v_of_rho     :      1.10s CPU      1.10s WALL (       6 calls)
     mix_rho      :      1.51s CPU      1.51s WALL (       5 calls)

     Called by c_bands:
     init_us_2    :      0.50s CPU      0.50s WALL (      11 calls)
     cegterg      :    882.01s CPU    882.01s WALL (       5 calls)

     Called by *egterg:
     h_psi        :    259.11s CPU    259.11s WALL (      17 calls)
     g_psi        :      9.02s CPU      9.02s WALL (      11 calls)
     cdiaghg      :    401.37s CPU    401.37s WALL (      16 calls)

     Called by h_psi:
     add_vuspsi   :     22.44s CPU     22.44s WALL (      17 calls)

     General routines
     calbec       :     17.25s CPU     17.25s WALL (      17 calls)
     fft          :      0.52s CPU      0.52s WALL (      66 calls)
     ffts         :      0.63s CPU      0.63s WALL (     117 calls)
     fftw         :    231.61s CPU    231.61s WALL (   10260 calls)
     davcio       :      4.72s CPU      4.72s WALL (       5 calls)

     Parallel routines
     fft_scatter  :     63.50s CPU     63.51s WALL (   10443 calls)
     ALLTOALL     :     10.66s CPU     10.67s WALL (   10252 calls)
     EXX routines

     PWSCF        : 17m42.94s CPU    17m42.94s WALL
```

```
     convergence NOT achieved after   5 iterations: stopping

     Writing output data file c8_atm213_k111.save

     init_run     :    119.48s CPU    119.48s WALL (       1 calls)
     electrons    :   1369.53s CPU   1369.53s WALL (       1 calls)

     Called by init_run:
     wfcinit      :     98.55s CPU     98.55s WALL (       1 calls)
     potinit      :      2.15s CPU      2.15s WALL (       1 calls)

     Called by electrons:
     c_bands      :   1289.41s CPU   1289.41s WALL (       5 calls)
     sum_band     :     56.06s CPU     56.06s WALL (       5 calls)
     v_of_rho     :      1.39s CPU      1.39s WALL (       6 calls)
     mix_rho      :      1.23s CPU      1.23s WALL (       5 calls)

     Called by c_bands:
     init_us_2    :      0.13s CPU      0.13s WALL (      11 calls)
     cegterg      :   1288.89s CPU   1288.89s WALL (       5 calls)

     Called by *egterg:
     h_psi        :    409.59s CPU    409.59s WALL (      17 calls)
     g_psi        :      2.35s CPU      2.35s WALL (      11 calls)
     cdiaghg      :    528.61s CPU    528.61s WALL (      16 calls)

     Called by h_psi:
     add_vuspsi   :     32.96s CPU     32.96s WALL (      17 calls)

     General routines
     calbec       :     31.22s CPU     31.22s WALL (      17 calls)
     fft          :      0.62s CPU      0.62s WALL (      66 calls)
     ffts         :      0.86s CPU      0.86s WALL (     117 calls)
     fftw         :    376.02s CPU    376.04s WALL (   82004 calls)
     davcio       :      6.38s CPU      6.38s WALL (       5 calls)

     Parallel routines
     fft_scatter  :     81.64s CPU     81.65s WALL (   82187 calls)

     PWSCF        : 24m57.48s CPU    24m57.48s WALL


    This run was terminated on:  12:25:36  12Oct2012
```